

BsP-SIC: Belief-selective Propagation-Aided SIC Detection for MIMO Systems

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Abstract—Tree-search detection such as K -best can achieve robust near-optimal performance but faces cubic complexity, limiting its application for massive MIMO scenarios. As the most simplified tree-search detection, successive interference cancellation (SIC) offers very low complexity but sacrifices performance benefits. To this end, we propose a belief-selective propagation-aided SIC (BsP-SIC) detector in this letter, which leverages the recently developed BsP algorithm to iteratively refine the SIC results. Two unexplored features: *forward soft-SIC* and *backward layered-BsP* are utilized to enhance performance without incurring excessive complexity. Numerical results demonstrate that the proposed BsP-SIC can achieve up to 1.5 dB performance gains over the traditional SIC detection at the bit error rate (BER) of 10^{-3} . Furthermore, compared to its message-passing counterparts, the BsP-SIC reduces 74%~90% complexity for 128×64 massive MIMO systems with 256-QAM.

Index Terms—MIMO detection, successive cancellation interference (SIC), belief-selective propagation (BsP).

I. INTRODUCTION

MULTIPLE-INPUT multiple-output (MIMO) detection encounters new challenges when faced with the IMT-2030 usage scenarios for the 6G vision [1]. Despite its optimal performance, the *maximum a posteriori* (MAP) detector suffers from prohibitive complexity, especially for large-scale MIMO arrays with high-order alphabets. The linear minimum mean squared error (LMMSE) detector [2] enjoys very low complexity, but its performance is far from optimal.

Achieving a well-balanced trade-off between complexity and performance, message-passing detectors (MPDs) have garnered extensive attention over the past decade. The expectation propagation (EP) detector [3] enjoys robust near-optimal performance for various MIMO scenarios but requires multiple matrix-inversion operations, hindering its implementation in massive MIMO systems. The large-MIMO approximate message passing algorithm (LAMA) detector [4] delivers near-optimal performance with low complexity for massive MIMO. However, it converges very slowly and suffers from aggravated performance degradation when the large system

limit is not satisfied. As an alternative MPD, the belief propagation (BP) detector [5] based on the factor graph (FG) shows great potential due to its near-optimal performance and hardware-friendly mechanism. As an approximate version of BP, the recently developed belief-selective propagation (BsP) [6] detector shows both performance and complexity merits by selecting messages with relatively high reliability for propagation. Furthermore, an implementation-ware BsP ASIC was designed in [7], demonstrating its hardware feasibility.

Tree-search detectors such as K -best [8] and sphere decoding (SD) [9] enjoy stable near-optimal performance but suffer from a complexity of cubic order, which is intractable for massive MIMO systems. As the most simplified tree-search detection, successive interference cancellation (SIC) shows low complexity and hardware-friendly merits but loses the performance benefits. To address this issue, we propose a BsP-aided SIC (BsP-SIC) detector in this letter, which utilizes the BsP algorithm to improve the performance of the SIC while maintaining its complexity merits. The main contributions are listed as follows.

- **Proposing BsP-SIC detection:** The BsP-aided SIC (BsP-SIC) detection is proposed for MIMO systems. By propagating messages with two directions: *forward soft-SIC* and *backward layered-BsP*, the BsP-SIC detector achieves up to 1.5 dB performance gains over the SIC at the bit error rate (BER) of 10^{-3} .
- **Achieving lower complexity:** Belief-selective propagation algorithm is adopted to deliver near-optimal performance while maintaining the complexity merits. Numerical results demonstrate that the BsP-SIC saves 74%–90% complexity over its message-passing counterparts in 128×64 256-QAM MIMO systems.
- **Facilitating faster convergence:** Layered message-updating strategy is utilized for the backward BsP algorithm to facilitate a faster convergence speed. Benefiting from this strategy, the proposed BsP-SIC can reach converged results with only 2 iterations for 128×64 256-QAM MIMO systems.

The remainder of this letter is organized as follows. Section II introduces the background. Section III provides details of the BsP-SIC detection. Numerical results are shown in Section IV. Finally, Section V summarizes this work.

Notations: Bold lowercase and uppercase letters represent vector and matrix, respectively. $\mathbf{H}_i$ denotes the i -th row vector of matrix $\mathbf{H}$. H_{ij} denotes the element of the i -th row and the j -th column of matrix $\mathbf{H}$. $\Re(\pm)$ and $\Im(\pm)$ denote the real and

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imaginary part of $\ddagger$, respectively.

II. PRELIMINARIES

A. MIMO System Model

We consider an uplink flat-fading MIMO system with N_t transmit antennas and N_r receive antennas. Then the received (RX) signal can be expressed as

$$\tilde{\mathbf{y}} = \tilde{\mathbf{H}}\tilde{\mathbf{s}} + \tilde{\mathbf{n}}, \quad (1)$$

where $\tilde{\mathbf{H}} \in \mathbb{C}^{N_r \times N_t}$ represents the channel matrix and each $\tilde{H}_{ij}$ follows complex Gaussian distribution as $\mathcal{CN}(0, 1)$ for Rayleigh channels, $\tilde{\mathbf{s}} \in \mathbb{C}^{N_t \times 1}$ and $\tilde{s}_j$ is chosen from the constellation $\mathcal{Q}$, and $\tilde{\mathbf{n}} \in \mathbb{C}^{N_r \times 1}$ is the noise vector with $\tilde{n}_i$ following $\mathcal{CN}(0, \sigma_n^2)$. For high-order high-dimensional MIMO scenarios, supposing $M = 2N_r$ and $N = 2N_t$, the real-domain system model is expressed as [10]

$$\mathbf{y} = \mathbf{H}\mathbf{s} + \mathbf{n}, \quad (2)$$

where $\mathbf{y} \in \mathbb{R}^{M \times 1} = \begin{bmatrix} \Re(\tilde{\mathbf{y}}) \\ \Im(\tilde{\mathbf{y}}) \end{bmatrix}$, $\mathbf{s} \in \mathbb{R}^{N \times 1} = \begin{bmatrix} \Re(\tilde{\mathbf{s}}) \\ \Im(\tilde{\mathbf{s}}) \end{bmatrix}$, $\mathbf{n} \in \mathbb{R}^{M \times 1} = \begin{bmatrix} \Re(\tilde{\mathbf{n}}) \\ \Im(\tilde{\mathbf{n}}) \end{bmatrix}$ and the real-domain channel matrix $\mathbf{H}$ can be derived by the mapping expression as

$$\mathbf{H} \in \mathbb{R}^{M \times N} = \begin{bmatrix} \Re(\tilde{\mathbf{H}}) & -\Im(\tilde{\mathbf{H}}) \\ \Im(\tilde{\mathbf{H}}) & \Re(\tilde{\mathbf{H}}) \end{bmatrix} \quad (3)$$

It is noted that the TX symbols in (2) are chosen from the real-domain constellation $\mathcal{Q} = \sqrt{\mathcal{Q}}$. For tree-search MIMO detection, QR decomposition often acts as the preprocessing operation. Applying QR decomposition for $\mathbf{H}$, the RX signal in (2) can be rewritten as

$$\hat{\mathbf{y}} = \mathbf{Q}^H \mathbf{y} = \mathbf{Q}^H \mathbf{Q} \mathbf{R} \mathbf{s} + \mathbf{Q}^H \mathbf{n} = \mathbf{R} \mathbf{s} + \hat{\mathbf{n}}, \quad (4)$$

where $\mathbf{Q} \in \mathbb{R}^{M \times N}$ is orthonormal matrix, and $\mathbf{R} \in \mathbb{R}^{N \times N}$ is upper triangular matrix. Utilizing the upper triangular feature of the $\mathbf{R}$ matrix, (4) can be spanned as

$$\begin{bmatrix} \hat{y}_1 \\ \hat{y}_2 \\ \vdots \\ \hat{y}_N \end{bmatrix} = \begin{bmatrix} R_{11} & R_{12} & \cdots & R_{1N} \\ 0 & R_{22} & \cdots & R_{2N} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & 0 & R_{NN} \end{bmatrix} \begin{bmatrix} s_1 \\ s_2 \\ \vdots \\ s_N \end{bmatrix} + \begin{bmatrix} \hat{n}_1 \\ \hat{n}_2 \\ \vdots \\ \hat{n}_N \end{bmatrix} \quad (5)$$

Based on (5), the SIC detection can be performed.

B. Belief Propagation Detection

The BP detection [5] was proposed based on the factor graph model, which involves M factor nodes (FNs) and N symbol nodes (SNs) denoted as f_i , $i \in \{1, 2, \dots, M\}$ and S_j , $j \in \{1, 2, \dots, N\}$, respectively. Soft messages are iteratively updated between FNs and SNs, until achieving the converged points or the maximum number of iterations. Define the message sent from i -th FN to j -th SN as β_{ij} , and it is computed by

$$\beta_{ij}(k) = \max_{\mathbf{s}: s_j = \mu_k} \left\{ \frac{-\|y_i - \mathbf{H}_i \mathbf{s}\|^2}{2\sigma_n^2} + \sum_{t=1, t \neq j}^N \alpha_{ti} \right\} - \max_{\mathbf{s}: s_j = \mu_1} \left\{ \frac{-\|y_i - \mathbf{H}_i \mathbf{s}\|^2}{2\sigma_n^2} + \sum_{t=1, t \neq j}^N \alpha_{ti} \right\}, \quad (6)$$

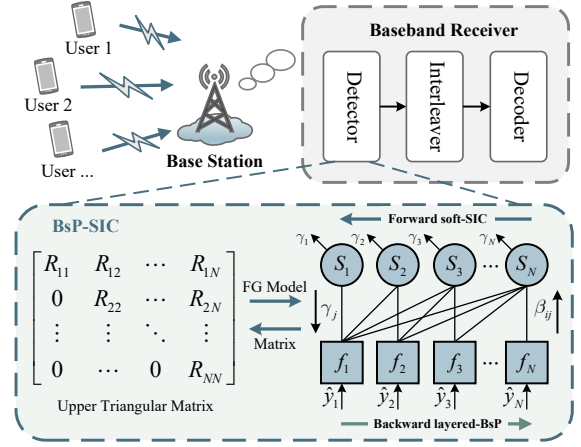


Fig. 1: MIMO system model with block diagram of the proposed BsP-SIC detection.

where μ_k is the k -th symbol of $\mathcal{Q}$, α_{ti} is the message sent from t -th SN to i -th FN in the preceding iteration, computed by

$$\alpha_{ti}(k) = \sum_{j=1, j \neq i}^M \beta_{jt}(k). \quad (7)$$

Note that all the messages discussed here are symbol log-likelihood ratios (LLRs). For instance, the message α_{ti} is defined as

$$\alpha_{ti}(k) = \ln \frac{P_t(\mu_k)}{P_t(\mu_1)}, \quad (8)$$

where $P_t(\mu_k)$ is the *a priori* probability of $s_t = \mu_k$.

III. THE PROPOSED BsP-SIC DETECTION

In this section, we introduce the methodology details of the proposed BsP-SIC detection, which involves two directions of message updating: *forward soft-SIC* and *backward layered-BsP*. Fig. 1 shows the FG model of the proposed BP-SIC detection after QR decomposition. Different from the conventional BP FGs, it circumvents the fully connected structure benefiting from the upper triangular matrix. In the forward message updating, the soft-SIC strategy is utilized to deliver soft messages, striving to achieve highly precise results. For the backward message updating, the layered-BsP algorithm is employed to refine the results and accelerate the convergence speed. More details of these two message-updating strategies are provided below.

A. Forward Soft-SIC

As the most simplified tree-search detection, SIC can be viewed as the K -best detector with $K = 1$. However, it can only output the hard-decision results of TX symbols, hindering the application of the backward layered-BsP. To this end, we develop a soft-output version of the SIC here, which can be easily derived from the QR decomposition-based MAP solution. Denoting γ_j^{MAP} as the MAP results of the j -th TX symbol, according to (4), it can be expressed as

$$\gamma_j^{\text{MAP}}(k) = \ln \frac{p(s_j = \mu_k | \mathbf{R}, \hat{\mathbf{y}})}{p(s_j = \mu_1 | \mathbf{R}, \hat{\mathbf{y}})}. \quad (9)$$

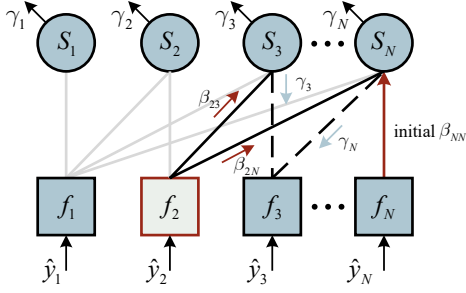


Fig. 2: Factor graph of the proposed BsP-SIC detection, where the 2-th FN is updating messages in the backward layered-BsP.

Utilizing the Bayesian rule and the Max-Log approximation, (9) can be derived as

$$\gamma_j^{\text{MAP}}(k) = \frac{1}{2\sigma_n^2} \left(\min_{\mathbf{s}: s_j = \mu_k} \|\hat{\mathbf{y}} - \mathbf{R}\mathbf{s}\|^2 - \min_{\mathbf{s}: s_j = \mu_1} \|\hat{\mathbf{y}} - \mathbf{R}\mathbf{s}\|^2 \right). \quad (10)$$

This MAP solution requires an exhaustive search ($|\mathcal{Q}|^N$) for possible TX symbols, resulting in prohibitive computational complexity. Fortunately, observing (5), γ_j^{MAP} can be computed successively, which only requires to search $|\mathcal{Q}|$ times of possible TX symbols. Therefore, (10) can be rewritten as

$$\gamma_j^{\text{MAP}}(k) = \frac{(\hat{y}_j - R_{jj}\mu_1 - \mathcal{E}_j)^2}{2\sigma_n^2} - \frac{(\hat{y}_j - R_{jj}\mu_k - \mathcal{E}_j)^2}{2\sigma_n^2}, \quad (11)$$

where $\mathcal{E}_j = \sum_{k=j+1}^N R_{jk}\hat{s}_k$ is the successive *multi-user interference* (MUI), with $\hat{s}_k$ representing the preceding estimated result of the k -th TX symbol. Actually, the message β_{jj} sent from the j -th FN to the j -th SN in the factor graph, which can be derived from (6), shares the same expression as (11). For convenience, we utilize β_{jj} to replace γ_j^{MAP} and define $\Phi_j = \hat{y}_j - \mathcal{E}_j$. Hence, (11) can be rewritten as

$$\begin{aligned} \beta_{jj}(k) &= \frac{(\Phi_j - R_{jj}\mu_1)^2}{2\sigma_n^2} - \frac{(\Phi_j - R_{jj}\mu_k)^2}{2\sigma_n^2} \\ &= [2\Phi_j R_{jj}(\mu_k - \mu_1) + R_{jj}^2(\mu_1^2 - \mu_k^2)]/2\sigma_n^2. \end{aligned} \quad (12)$$

After computing the message β_{jj} , the message γ_j in Fig. 1 needs to be updated, illustrated as

$$\gamma_j(k) = \sum_{i=1}^j \beta_{ij}(k). \quad (13)$$

With $\gamma_j(k)$, $\hat{s}_j$ is computed as

$$\hat{s}_j = \sum_{k=1}^{|\mathcal{Q}|} P_j(\mu_k) \mu_k \quad (14)$$

where $P_j(\mu_k) = \exp(\gamma_j(k) - \max\{\gamma_j\})$. Here note that $\mathcal{E}_N$ equals *zero* when computing the message β_{NN} . In the following rounds of forward message updating, β_{NN} will not be updated, since it no longer provides extrinsic information. However, the message γ_N will be updated as (13) for each round of forward message updating. Alongside the unchanged β_{NN} , the messages β_{iN} , $i = \{1, 2, \dots, N-1\}$ carrying extrinsic information are updated in the backward layered-BsP, which will be detailed in the next subsection.

Algorithm 1: BsP-SIC Detection Algorithm

Input : $\mathbf{H}$, $\mathbf{y}$, $\mathcal{Q}$, $I_{\max}$.
Output : symbol LLRs γ .

- 1 Initialization: $\hat{\mathbf{y}} = \mathbf{R}\mathbf{s} + \hat{\mathbf{n}}$; $\triangleright$ utilizing QR decomposition
- 2 $\forall \mu_k \in \mathcal{Q}$: $\beta_{NN}(k) = [2\hat{y}_N R_{NN}(\mu_k - \mu_1) + R_{NN}^2(\mu_1^2 - \mu_k^2)]/2\sigma_n^2$; $\triangleright$ acquire the initial β_{NN} message
- 3 **for** Iter = 1 $\rightarrow$ $I_{\max}$ **do**
- 4 **for** $j = N-1 \rightarrow 1$ **do** $\triangleright$ soft-SIC
- 5 $\forall \mu_k \in \mathcal{Q}$: $\gamma_{j+1}(k) = \sum_{i=1}^{j+1} \beta_{ij+1}(k)$;
 $P_{j+1}(\mu_k) = \exp(\gamma_{j+1}(k) - \max\{\gamma_{j+1}\})$;
 $\hat{s}_{j+1} = \sum_{k=1}^{|\mathcal{Q}|} P_{j+1}(\mu_k) \mu_k$;
- 6 $\mathcal{E}_j = \sum_{k=j+1}^N R_{jk}\hat{s}_k$; $\Phi_j = \hat{y}_j - \mathcal{E}_j$;
- 7 $\forall \mu_k \in \mathcal{Q}$: $\beta_{jj}(k) = [2\Phi_j R_{jj}(\mu_k - \mu_1) + R_{jj}^2(\mu_1^2 - \mu_k^2)]/2\sigma_n^2$.
- 8 **for** $i = 1 \rightarrow N-1$ **do** $\triangleright$ layered-BsP
- 9 $\forall (j \in [i, N], \mu_k \in \mathcal{Q})$: $\gamma_j(k) = \sum_{m=1}^j \beta_{mj}(k)$;
 $\hat{s}_j^{\max} = \arg \max_{\mu_k} \gamma_j(k)$;
- 10 $\forall (j \in [i+1, N], t \in [i, N] (t \neq j), \mu_k \in \mathcal{Q})$:
 $\phi_{ij} = \hat{y}_i - \sum_{t=i, t \neq j}^N R_{it}\hat{s}_t^{\max}$; $\beta_{ij}(k) = \frac{1}{2\sigma_n^2} (\phi_{ij} - R_{ij}\mu_1)^2 - \frac{1}{2\sigma_n^2} (\phi_{ij} - R_{ij}\mu_k)^2$.
- 11 Output: $\gamma_j(k) = \sum_{i=1}^j \beta_{ij}(k)$.

B. Backward Layered-BsP

Although the forward soft-SIC enjoys relatively low complexity, it suffers from aggravated performance degradation due to successive error accumulation. To address this problem, we develop the BsP algorithm to refine the SIC's results when updating the backward message. The BsP algorithm is proposed in our previous work [6], which only utilizes messages with relatively large reliability for updating. Moreover, the layered message-updating technique is applied to accelerate the convergence speed. Fig. 2 represents the factor graph of the proposed BsP-SIC detector, where f_2 is updating messages for the backward layered-BsP. After receiving the newly computed β messages, SNs will update γ messages immediately to enable the message updating of the next FN. The β_{ij} is computed by

$$\begin{aligned} \beta_{ij}(k) &= -\frac{1}{2\sigma_n^2} (\hat{y}_i - \sum_{t=i, t \neq j}^N R_{it}s_t^{\max} - R_{ij}\mu_k)^2 \\ &\quad + \frac{1}{2\sigma_n^2} (\hat{y}_i - \sum_{t=i, t \neq j}^N R_{it}s_t^{\max} - R_{ij}\mu_1)^2, \end{aligned} \quad (15)$$

where $s_t^{\max}$ is the most reliable constellation point acquired by performing $\max\{\gamma_t\}$. For the readers' better understanding,

Alg. 1 summarises the processing details of the proposed BsP-SIC detection algorithm with $I_{\max}$ iterations. Before the message-updating iterations, the message β_{NN} is first computed as the initial information, as shown in Fig. 2. Then, the proposed BsP-SIC detector iteratively performs forward soft-SIC and backward layered-BsP for message updates. Finally, when the maximum iteration number is achieved, each symbol node will output the γ messages by accumulating all of its received β messages.

IV. NUMERICAL RESULTS

In this section, we provide numerical results to demonstrate the effectiveness of the proposed BsP-SIC detection. Alongside the SIC detection, the BsP-SIC is compared with the other well-established detectors, involving the LMMSE detection [2], the LAMA detection [4], the BsP detection [6], and the EP detection [3]. For the BsP detection, the most simplified configuration $\mathcal{B}(1, 1)$ is utilized. The damping factors of the EP and LAMA detection are set as $\delta = 0.33$.

A. Performance Evaluation

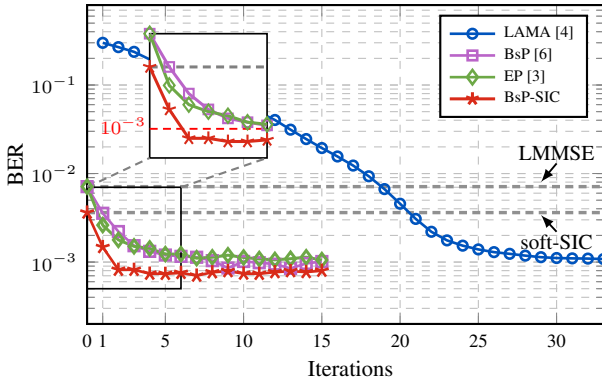


Fig. 3: Convergence results of the proposed BsP-SIC in 128×64 256-QAM Rayleigh fading MIMO system at SNR=20 dB.

Without loss of generality, we first provide results of BER vs. Iterations for the proposed BsP-SIC and other detectors in a 128×64 256-QAM MIMO system to demonstrate its convergence performance. The iteration 0 here denotes the results of LMMSE initialization (for BsP and EP) and soft-SIC (for BsP-SIC). From Fig. 3 it can be observed that the LAMA detection requires nearly 30 iterations to converge. The BsP and the EP detection require 5 iterations to achieve converged results. The proposed BsP-SIC enjoys the fastest convergence speed, with only 2 iterations reaching converged results. Fig. 4(a) shows the BER performance of the proposed BsP-SIC detector in 8×4 256-QAM MIMO systems under Rayleigh channels. It is observed that the proposed BsP-SIC earns about 1.0 dB performance gains over the SIC detection at $\text{BER}=10^{-3}$. Moreover, the proposed BsP-SIC achieves about 1.1 dB and 1.5 dB performance gains at $\text{BER}=10^{-3}$ over the BsP and the LAMA detection. Compared with the near-optimal EP and the optimal SESD detection [11], the proposed BsP-SIC suffers from slight performance loss at high SNR regions. To demonstrate its applications in practical scenarios,

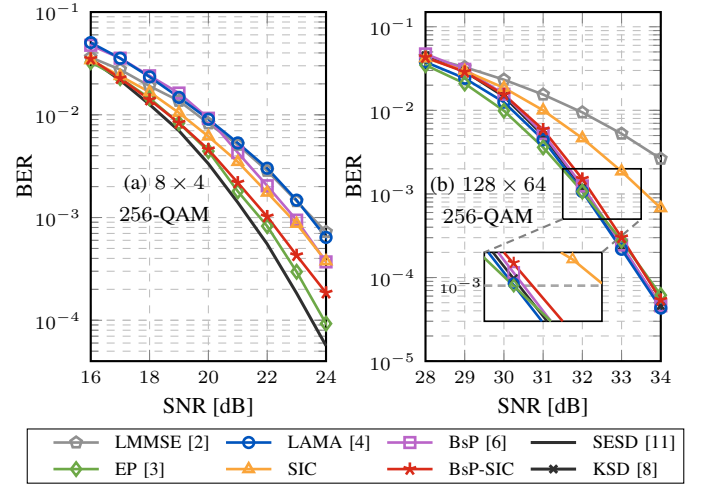


Fig. 4: BER performance of the proposed BsP-SIC detector in: (a) 8×4 MIMO systems under Rayleigh channels and (b) 128×64 MIMO systems under WINNER-II channels.

Fig. 4(b) also reports the BER results of the proposed BsP-SIC in 128×64 256-QAM MIMO scenarios under WINNER-II channels [12] and utilizing least square (LS) channel estimation. The K -best sphere decoding (KSD) [8] with $K = 32$ is involved here to show the near-optimal performance. It can be seen that the proposed BsP-SIC detection delivers 1.5 dB performance gains at $\text{BER}=10^{-3}$ over the SIC detection. In addition, the BsP-SIC detection achieves tight BER results compared with the near-optimal KSD.

B. Complexity Analysis

Table I summarizes the complexity of the proposed BsP-SIC and other message-passing counterparts. For a fair comparison, all operations involving multiplications, additions, and exponentiations of each detector are summarized. The complexity of the initialization for each detector is also considered. Derived from Alg. 1, the forward message updating of the proposed BsP-SIC requires $(N-1)|Q|$, $\frac{1}{2}N(N-1)$, and $2N|Q|$ multiplications to compute the $\hat{s}_j$, $\mathcal{E}_j$, and $R_{ii}\mu_k$, respectively. The backward message updating needs $\frac{1}{2}N(N-1)$ and $N(N-1)|Q|$ multiplications to compute the ϕ_{ij} and β_{ij} , respectively. Accumulating the multiplications above and attention to the highest order, the multiplications of the proposed BsP-SIC with $I_{\max}$ iterations can be expressed as $\mathcal{O}(N^2|Q|I_{\max})$. Similarly, the additions of the proposed BsP-SIC with $I_{\max}$ can be represented as $\mathcal{O}(\frac{1}{4}N^2|Q|^2I_{\max})$, in which the backward BsP involves $\frac{1}{4}N(N-1)|Q|(|Q|-1)$ comparisons¹ to find the $\hat{s}_j^{\max}$, serving as the highest order of the additions. In addition, the proposed BsP-SIC also requires $N|Q|$ exponentiations to update the symbol *a priori* probability P_j for each iteration. Therefore, it is written as $\mathcal{O}(N|Q|I_{\max})$ with $I_{\max}$ iterations. The complexity of the LAMA, the EP, and the BsP detection can be drawn from [4], [3], and [6], respectively.

¹We note that a comparison is implemented by performing a subtraction, which has the same complexity as an addition.

TABLE I: Complexity Comparisons of Different Detectors

Detectors	Multiplications	Additions	Exponentiations
LAMA [4]	$\mathcal{O}((2NM + 4N \mathcal{Q})I_{\max})$	$\mathcal{O}((2NM + 4N \mathcal{Q})I_{\max})$	$\mathcal{O}(N \mathcal{Q} I_{\max})$
EP [3]	$\mathcal{O}((\frac{2}{3}N^3 + 4N \mathcal{Q})I_{\max})$	$\mathcal{O}((\frac{2}{3}N^3 + 5N \mathcal{Q})I_{\max})$	$\mathcal{O}(N \mathcal{Q} I_{\max})$
BsP [6]	$\mathcal{O}(2MN \mathcal{Q} I_{\max})$	$\mathcal{O}(\frac{1}{2}MN \mathcal{Q} ^2I_{\max})$	0
BsP-SIC	$\mathcal{O}(N^2 \mathcal{Q} I_{\max})$	$\mathcal{O}(\frac{1}{4}N^2 \mathcal{Q} ^2I_{\max})$	$\mathcal{O}(N \mathcal{Q} I_{\max})$

¹ The BsP detection utilize the LMMSE initialization, requiring $\mathcal{O}(\frac{2}{3}N^3)$ multiplications and additions.

² The proposed BsP-SIC detection needs to perform QR decomposition, which has $\mathcal{O}(\frac{2}{3}N^3)$ multiplications and additions.

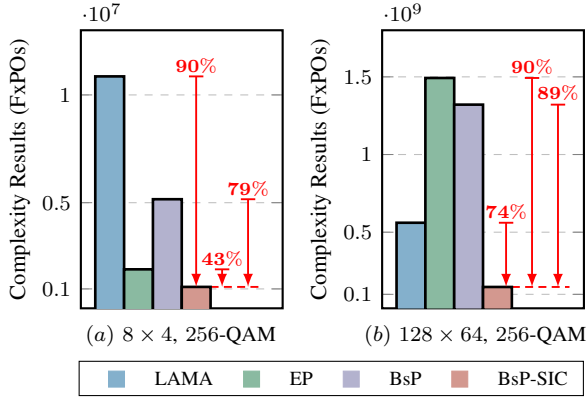


Fig. 5: FxPOs of different detectors in (a) 8×4 , 256-QAM and (b) 128×64 , 256-QAM MIMO systems.

Alongside the complexity expression in Table I, a more vivid complexity comparison is made by normalizing the multiplications, additions, and exponentiations with Fixed-Point Operations (FxPOs). Assuming k and m are the bit width of the adder and the exponential function in hardware implementation, the FxPOs can be computed by [7]

$$\text{FxPOs} = k^2 \text{Mul.} + k \text{Add.} + mk^2 \text{Exp.}, \quad (16)$$

where $m = 8$ and $k = 14$ are often utilized for the quantization of different MIMO detectors [7, 13]. Fig. 5 reports the FxPOs results of the proposed BsP-SIC and the other detectors in the medium-scale 8×4 and large-scale 128×64 MIMO systems with 256-QAM modulation. Several observations can be drawn from Fig. 5. First, compared to the LAMA detection, the proposed BsP-SIC achieves 90% and 74% complexity reduction in the medium and large-scale MIMO scenarios, respectively. Note that the LAMA also suffers from aggravated performance degradation when applied with medium-scale MIMO arrays. Second, compared to the EP detection, the proposed BsP-SIC enjoys an order-lower complexity, saving 90% FxPOs in the large-scale MIMO scenario. Finally, benchmarking with the recently emerged BsP detection, the proposed BsP-SIC succeeds in achieving 79%–89% FxPOs reduction for the medium and large-scale MIMO scenarios.

V. CONCLUSION

The BsP-SIC detection is developed in this letter, which utilizes the belief-selective strategy to enhance the performance

of the SIC detection. The proposed BsP-SIC achieves up to 1.5 dB performance gains at $\text{BER}=10^{-3}$ over the SIC detection. Compared to the other well-established message-passing counterparts, the proposed BsP-SIC delivers comparable near-optimal BER performance but achieves 74%–90% complexity reduction for the 128×64 256-QAM MIMO system. Future work will be directed towards the hardware implementation of the BsP-SIC detector to demonstrate its feasibility.

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